

MEDIATEK

MT8676 Yocto GPS User Manual

Version: 1.0
Release date: 2024-08-12

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Version History

Version	Date	Author	Description
1.0	2024-08-12	Neo.Sun	Official release

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1 GPS

1.1 Overview

1.1.1 Brief Introduction

This section introduces the basic functions of the MT8676 GPS and the solutions to common problems.

1.1.2 GPS Abbreviations

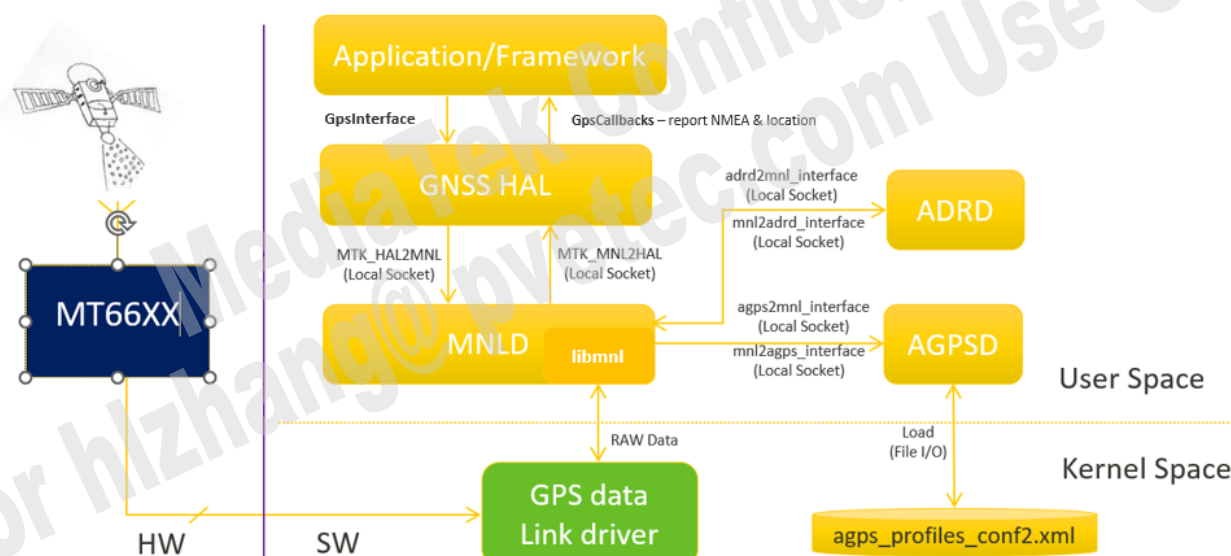
Table 1-1. Abbreviations

Abbreviation	Explanation
COLD start	With time-assisted information, the end user will not encounter this scenario.
FULL start	Without any auxiliary information, equivalent to the scenario where an end user uses a positioning application for the first time after purchasing a mobile phone.
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
Hot start	With all auxiliary information, the end user's current positioning occurs less than 2 to 4 hours after the last positioning.
NMEA	National Marine Electronics Association. A communication protocol used for data exchange between marine electronic devices, widely applied in GPS/GNSS receiver data output.
TTF	Time To First Fix. The time required for a navigation device to successfully acquire the first valid positioning data from startup.
WARM start	With time and location-assisted information, the end user's current positioning occurs more than 2 to 4 hours after the last positioning.

1.2 Architecture/Process Overview

1.2.1 GPS Architecture

MT8676 GPS architecture:



MNLD: MediaTek Navigation Library Daemon

Figure 1-1. GPS architecture

1.2.2 Common Flow

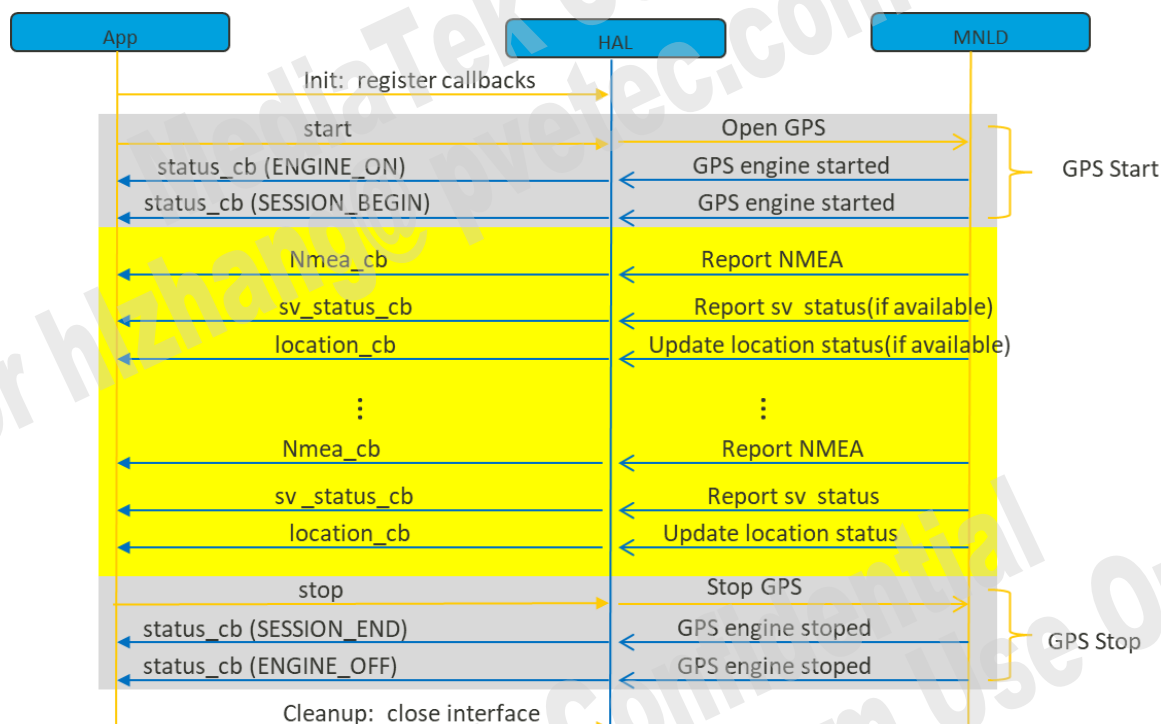


Figure 1-2. Common flow

1.3 Configuration/Customization Guideline

1.3.1 Fix Rate Configuration

Fix Rate refers to the rate at which GNSS reports position information. The fixed rates currently supported by MT8676 include: 1Hz, 2Hz, 5Hz, and 10Hz, with the default configuration being 1Hz output. The method to change the fixed rate is as follows:

Method1: Modify the code to configure the `fix_interval` parameter. `fix_interval = 100` corresponds to 10Hz; `fix_interval = 1000` corresponds to 1Hz.



Figure 1-3. Fix Rate configuration

Method2: Dynamically modify the configuration file, which will take effect after restarting GNSS. Command: `echo fix_interval=1000 >> /etc/gnss/mn1.prop /Set to 1Hz, effective after GPS restart.`

1.3.2 Gnss Mode Configuration

MT8676 supports GPS + GLONASS + Galileo + BeiDou multi-satellite navigation positioning systems. The default configuration for `gnssopmode` is set to `MTK_CONFIG_GPS_GLONASS_BEIDOU_GALILEO_NAVIC` (the default configuration offers the best GNSS performance, and it is recommended to use the default configuration).

```

885 typedef enum
886 {
887     MTK_CONFIG_GPS_GLONASS = 0,
888     MTK_CONFIG_GPS_BEIDOU,
889     MTK_CONFIG_GPS_GLONASS_BEIDOU,
890     MTK_CONFIG_GPS_ONLY,
891     MTK_CONFIG_BEIDOU_ONLY,
892     MTK_CONFIG_GLONASS_ONLY,
893     MTK_CONFIG_GPS_GLONASS_BEIDOU_GALILEO,
894     MTK_CONFIG_GPS_GALILEO,
895     MTK_CONFIG_GPS_GLONASS_GALILEO,
896     MTK_CONFIG_GALILEO_ONLY,
897     MTK_CONFIG_GPS_GLONASS_BEIDOU_GALILEO_NAVIC,
898     MTK_CONFIG_GNSS_END
899 } MTK_GNSS_CONFIGURATION;

```

Figure 1-4. GNSS configuration

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 Log Related Issues

- What logs do MediaTek engineers need for problem analysis?

/data/debuglogger/mobilelog and NMEA log

The mobilelog is enabled by default. To enable the NMEA log, you need to create the `/etc/gnss/mnl.prop` file and write the following into it:

```
debug.dbg2file=1
```

```
debug.filename=/data/debuglogger/gpsdebug.log
```

Save and reboot, and the NMEA log will appear in `/data/debuglogger/gpsdebug.log`.

1.4.2 Testing Related Issues

- Before testing, it is necessary to check whether there is a satellite signal and whether it is in an open sky environment.

To test GNSS satellite search or positioning functions, the signal needs to be in an open sky environment, such as an open outdoor area or a laboratory with a signal amplifier.

The prerequisite for successful positioning is having more than 6 satellites with a CNR of 40~43 dBm. ---> It is crucial to pay attention to this when testing GNSS. If you are unsure whether the current signal environment meets the requirements, place a comparison device in the same environment for comparison.

- How to test the TTFF (Time To First Fix) for different startup modes such as FULL start, WARM start, COLD start, and HOT start?

You can use `mnld_test`.

Start test(open gps): `mnld_test -h`

```
sh-3.2# mnld_test -h
mnld_test -h
mnld_test -h
MNLD client test :
-----
The command to start GNSS test:
mnld_test start [start type] [restart times] [restart interval]
[start type]:
h/H: hot start;
w/W: Warm start;
c/C: Cold start;
f/F: Full start
[restart times]: integer value, range is 0-1000, default is 0(no restart, always on).
[restart interval]: integer value, range is 0-3600, the unit is second; the default value is 60
-----
The command to stop GNSS test:
Ctrl+C or kill -9 `pidof mnld_test`
-----
The command to update network status:
mnld_test network [type] [roaming]
[type]: wifi, mobile, disable
[roaming]: roaming, the mobile network is in roaming state
sh-3.2#
```

Figure 1-5. Mnld test

According to the instructions, for example, if you need to test a cold start, you can use the following command:

```
mnld_test start c &
```

You can enter the following command to output the log to the serial port, where you can see TTFF and other information:

```
journalctl -f --no-tail -o short-precise |grep mnldtest
```

1.4.3 GNSS Path

Recipes & source code:

```
meta/meta-mediatek/recipes-connectivity/mnld  
src/connectivity/gps/4.0
```

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